

Introduction to Function Spaces

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Introduction

I prepared these lecture notes for my students to have a concise and coherent reference for the material covered in the course. The topics of L^p spaces, the Fourier transform, tempered distributions, and Sobolev spaces form the analytic toolkit behind much modern real analysis and the theory of partial differential equations.

The notes draw inspiration from lectures I once attended myself, particularly those of my former supervisor Valery Serov [5], as well as from other excellent courses given around Finland (such as Mikko Salo's course on Function Spaces [4]), and from several classical textbooks. My goal has simply been to gather the essential material in one place and present it in a way that I find natural and, hopefully, useful for students encountering these ideas for the first time.

Suggested background The student attending this course is expected to be familiar with the following topics:

- Basic concepts from complex analysis
- The notion of complete normed spaces (Banach spaces)
- Some familiarity with measure spaces and the Lebesgue integral
- Fluency in multivariable calculus

Although it is certainly possible to follow the course with limited prior knowledge of these topics, some definitions and techniques may sometimes seem to appear “out of thin air.”

Motivation

The Fourier transform is one of the central tools of modern analysis. It allows us to decompose functions into their frequency components, and thus provides a bridge between the “time domain” and the “frequency domain.” This duality lies at the heart of applications ranging from partial differential equations to signal processing, quantum mechanics, and image analysis.

In the early 19th century, Fourier introduced the idea of representing a function f on the interval $[-\pi, \pi]$ by a trigonometric series

$$f(x) \sim \sum_{k \in \mathbb{Z}} c_k e^{ikx}, \quad c_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-ikx} dx.$$

This is now known as the *Fourier series*. The coefficients c_k measure how much of the frequency k is present in the signal.

Passing to functions on the whole real line, it becomes natural to replace the discrete sum by an integral. This leads to the Fourier transform

$$\widehat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ix\xi} dx,$$

which represents f as a superposition of plane waves $e^{-ix\xi}$ with frequency parameter $\xi \in \mathbb{R}$.

The Fourier transform has three fundamental features:

- **Algebraic:** It converts convolution into multiplication,

$$\widehat{f * g}(\xi) = \widehat{f}(\xi) \widehat{g}(\xi).$$

This makes it a natural tool for studying linear operators with translation-invariant kernels.

- **Analytic:** It provides precise information on regularity and decay. For instance, smoothness of f corresponds to rapid decay of $\widehat{f}$.
- **Geometric:** It connects localization in space with spread in frequency, leading to results such as the *uncertainty principle*.

Organization of these notes

We begin by recalling the basic functional analytic framework of L^p -spaces, before introducing the Schwartz space and the Fourier transform. We then extend the Fourier transform to L^p -functions, and develop the language of Sobolev spaces for measuring smoothness. Finally, we prove a version of the Sobolev embedding theorem, which connects the differentiability and integrability of functions.

Chapter 1

L^p -spaces

How can one measure the *size of a function*? You might already be familiar with some notions of size, such as

- Pointwise bounds: $|f(x)| \leq C$ for some $C > 0$ and all $x \in \mathbb{R}^n$.
- Area $\int |f(x)| dx$ under the graph of f .

Here the integral of $|f|$ does measure the total *mass* of the function f , but it does not distinguish whether the mass is spread out or localized. To give an answer to this question, we define the Lebesgue spaces L^p .

Let $(X, \mathcal{M}, \mu)$ be a measure space. For $1 \leq p < \infty$, we define the $\mathcal{L}^p$ -spaces by

$$\mathcal{L}^p(X) = \left\{ f : X \rightarrow \mathbb{C} \mid \|f\|_{L^p} := \left(\int_X |f(x)|^p d\mu(x) \right)^{1/p} < \infty \right\}.$$

For $p = \infty$, we set

$$\mathcal{L}^\infty(X) = \{f : X \rightarrow \mathbb{C} \mid \|f\|_{L^\infty} := \operatorname{ess\,sup}_{x \in X} |f(x)| < \infty\}.$$

On this course, we will only work on the measure space $(\mathbb{R}^n, \mathcal{M}, dx)$, where $\mathcal{M}$ is the Lebesgue σ -algebra and dx the Lebesgue measure.

We will next construct the *Lebesgue spaces* $L^p(\mathbb{R}^n)$ by identifying functions as follows: $f, g : X \rightarrow \mathbb{C}$ are considered *equal* if they agree almost everywhere, that is,

$$f(x) = g(x) \quad \text{for } \mu\text{-almost every } x \in X.$$

This quotient construction ensures that the L^p -norm is a true norm (vanishing only on the zero element) and that $L^p(X)$ becomes a normed vector space. Without taking this identification, one only has a *seminorm*, since $\|f\|_{L^p} = 0$ can occur for a nonzero function on a set of measure zero. Precisely, let

$$\mathcal{N} := \{f : \mathbb{R}^n \rightarrow \mathbb{C} \mid f \text{ measurable and } f = 0 \text{ a.e.}\}.$$

Then, the Lebesgue spaces are defined as the quotient spaces

$$L^p(\mathbb{R}^n) := \mathcal{L}^p(\mathbb{R}^n)/\mathcal{N}.$$

Thus, two cosets $[f], [g] \in L^p(\mathbb{R}^n)$ are equal iff $f = g$ a.e.

The distinction between the spaces $\mathcal{L}^p$ of *functions* and the spaces L^p of *equivalence classes* is short-lived, and henceforth we will omit the coset brackets, and simply work with (representative) functions $f \in L^p(\mathbb{R}^n)$.

Exercise 1.0.1. Let $\chi_A : X \rightarrow \mathbb{C}$ be the *characteristic function* of the set $A \subset X$:

$$\chi_A(x) = \begin{cases} 1, & x \in A \\ 0, & \text{otherwise.} \end{cases}$$

Let $\varepsilon > 0$ and $1 \leq p, q \leq \infty$, and calculate $\|\varepsilon^{-1/p} \chi_{[0, \varepsilon]}\|_{L^q([0, 1])}$. What happens as $\varepsilon \rightarrow 0^+$, when $p < q$, $p = q$, or $p > q$?

Theorem 1.0.2 (Hölder's inequality). *Let $1 \leq p, q \leq \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$ (the number q is called the Hölder conjugate of p). If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, then*

$$\int_{\mathbb{R}^n} |f(x)g(x)| dx \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Proof. **Case** $p = 1, q = \infty$. Since $g \in L^\infty$ we have $|g(x)| \leq \|g\|_{L^\infty}$ a.e., hence¹

$$\int_{\mathbb{R}^n} |fg| \leq \|g\|_{L^\infty} \int_{\mathbb{R}^n} |f| = \|f\|_{L^1} \|g\|_{L^\infty}.$$

The case $p = \infty$ and $q = 1$ is exactly the same.

Case $1 < p, q < \infty$. We assume $f, g \geq 0$ for the inequality below (the general case follows by applying the inequality to $|f|, |g|$).

We first prove Young's inequality for products: for $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

This follows by noting that $f(x) = e^x$ is convex ($f''(x) = e^x > 0$) and therefore

$$\begin{aligned} ab &= e^{\log(ab)} = e^{\log(a) + \log(b)} = e^{\frac{1}{p} \log(a^p) + \frac{1}{q} \log(b^q)} \\ &= f\left(\frac{1}{p} \log(a^p) + \frac{1}{q} \log(b^q)\right) \leq \frac{1}{p} f(\log(a^p)) + \frac{1}{q} f(\log(b^q)) = \frac{a^p}{p} + \frac{b^q}{q}. \end{aligned}$$

Apply this pointwise with

$$a = \frac{f(x)}{\|f\|_{L^p}}, \quad b = \frac{g(x)}{\|g\|_{L^q}},$$

¹When no confusion can arise, we often omit the explicit dependence on the variables inside integrals. Thus $\int f dx = \int f(x) dx$.

(which are well-defined a.e. provided $\|f\|_{L^q}, \|g\|_{L^q}$ are nonzero; if one of the them is zero the inequality is trivial) to obtain

$$\frac{|f(x)g(x)|}{\|f\|_{L^p}\|g\|_{L^q}} \leq \frac{1}{p} \left(\frac{|f(x)|}{\|f\|_{L^p}} \right)^p + \frac{1}{q} \left(\frac{|g(x)|}{\|g\|_{L^q}} \right)^q.$$

Integrate both sides over $\mathbb{R}^n$ and use homogeneity of $\|\cdot\|_{L^q}$:

$$\frac{\int_{\mathbb{R}^n} |fg|}{\|f\|_{L^p}\|g\|_{L^q}} \leq \frac{1}{p} \frac{\|f\|_{L^p}^p}{\|f\|_{L^p}^p} + \frac{1}{q} \frac{\|g\|_{L^q}^q}{\|g\|_{L^q}^q} = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying through by $\|f\|_{L^p}\|g\|_{L^q}$ yields the desired inequality. $\square$

Exercise 1.0.3. Prove that $\|fg\|_{L^1} = \|f\|_{L^p}\|g\|_{L^q}$ if and only if $|f|^p = \lambda|g|^q$ a.e.

Corollary 1.0.4 (Cauchy-Schwarz). *For $f, g \in L^2(\mathbb{R}^n)$, we have*

$$\left| \int_{\mathbb{R}^n} f(x)\overline{g(x)} dx \right| \leq \|f\|_{L^2} \|g\|_{L^2}.$$

Hence $L^2(\mathbb{R}^n)$ is an inner product space with

$$\langle f, g \rangle_{L^2} = \int_{\mathbb{R}^n} f(x)\overline{g(x)} dx.$$

The inequality for sums was published by Augustin-Louis Cauchy (1821). The corresponding inequality for integrals was published by Viktor Bunyakovsky (1859) and Hermann Schwarz (1888); it is for this reason the inequality is sometimes called the Cauchy-Bunyakovsky-Schwarz inequality.

Cauchy-Schwarz inequality only applies to $L^2(\mathbb{R}^n)$ -spaces and implies directly the triangle inequality. For general $p \neq 2$, we need to prove separately, that $\|\cdot\|_{L^p}$ satisfies the triangle inequality. This is the content of

Theorem 1.0.5 (Minkowski inequality). *For $1 \leq p \leq \infty$ and $f, g \in L^p(\mathbb{R}^n)$, we have*

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

Thus $(L^p(\mathbb{R}^n), \|\cdot\|_{L^p})$ is a normed vector space.

Proof. The cases $p = 1$ and $p = \infty$ are immediate:

Case $p = 1$. By the triangle inequality for absolute value,

$$\begin{aligned} \|f + g\|_{L^1} &= \int_{\mathbb{R}^n} |f(x) + g(x)| dx \\ &\leq \int_{\mathbb{R}^n} |f(x)| dx + \int_{\mathbb{R}^n} |g(x)| dx \\ &= \|f\|_{L^1} + \|g\|_{L^1}. \end{aligned}$$

Case $p = \infty$. By definition of the essential supremum,

$$\begin{aligned}\|f + g\|_{L^\infty} &= \operatorname{ess\,sup}_x |f(x) + g(x)| \\ &\leq \operatorname{ess\,sup}_x |f(x)| + \operatorname{ess\,sup}_x |g(x)| \\ &= \|f\|_{L^\infty} + \|g\|_{L^\infty}.\end{aligned}$$

Case $1 < p < \infty$. Let p' be the conjugate exponent, $1/p + 1/p' = 1$. Write

$$\begin{aligned}\int_{\mathbb{R}^n} |f + g|^p dx &= \int_{\mathbb{R}^n} |f + g| |f + g|^{p-1} dx \\ &\leq \int_{\mathbb{R}^n} |f| |f + g|^{p-1} dx + \int_{\mathbb{R}^n} |g| |f + g|^{p-1} dx.\end{aligned}$$

Apply Hölder's inequality to each term with exponents p and p' :

$$\int_{\mathbb{R}^n} |f| |f + g|^{p-1} dx \leq \|f\|_{L^p} \| |f + g|^{p-1} \|_{L^{p'}},$$

and similarly for the term with g . Observe that

$$\| |f + g|^{p-1} \|_{L^{p'}} = \left(\int_{\mathbb{R}^n} (|f + g|^{p-1})^{p'} dx \right)^{1/p'} = \left(\int_{\mathbb{R}^n} |f + g|^p dx \right)^{1/p'}.$$

Hence

$$\int_{\mathbb{R}^n} |f + g|^p dx \leq (\|f\|_{L^p} + \|g\|_{L^p}) \left(\int_{\mathbb{R}^n} |f + g|^p dx \right)^{1/p'}.$$

Set $A := \left(\int_{\mathbb{R}^n} |f + g|^p dx \right)^{1/p} = \|f + g\|_{L^p}$. Since $1/p' = 1 - 1/p$ we have

$$A^p \leq (\|f\|_{L^p} + \|g\|_{L^p}) A^{p/p'} = (\|f\|_{L^p} + \|g\|_{L^p}) A^{p-1}.$$

If $A = 0$ the claimed inequality is trivial, otherwise divide by A^{p-1} to get

$$A \leq \|f\|_{L^p} + \|g\|_{L^p},$$

which is the desired Minkowski inequality.

Finally, combining Minkowski with the homogeneity and the fact that $\|f\|_{L^p} = 0$ iff $f = 0$ a.e., we see that $\|\cdot\|_{L^p}$ is indeed a (genuine) norm on $L^p(\mathbb{R}^n)$. $\square$

1.1 Completeness

Theorem 1.1.1 (Riesz-Fischer). *For $1 \leq p \leq \infty$, the space $L^p(\mathbb{R}^n)$ is complete, hence a Banach space.*

Proof. **Case $1 \leq p < \infty$.** Let $(f_k)_{k \geq 1}$ be a Cauchy sequence in $L^p(\mathbb{R}^n)$. We must find $f \in L^p(\mathbb{R}^n)$ such that $\|f_k - f\|_{L^p} \rightarrow 0$.

Because (f_k) is Cauchy, there exists a subsequence (f_{k_j}) satisfying

$$\|f_{k_{j+1}} - f_{k_j}\|_{L^p} \leq 2^{-j}, \quad j \geq 1.$$

Define the partial sums

$$g_n(x) := \sum_{j=1}^n |f_{k_{j+1}}(x) - f_{k_j}(x)|, \quad g(x) := \sum_{j=1}^{\infty} |f_{k_{j+1}}(x) - f_{k_j}(x)|.$$

By the triangle inequality (Minkowski's inequality) in L^p ,

$$\|g_n\|_{L^p} \leq \sum_{j=1}^n \|f_{k_{j+1}} - f_{k_j}\|_{L^p} \leq \sum_{j=1}^n 2^{-j} \leq 1.$$

Hence (g_n) is an increasing sequence of nonnegative functions bounded in L^p , so by Fatou's lemma,

$$\int_{\mathbb{R}^n} |g(x)|^p dx \leq \liminf_{n \rightarrow \infty} \int_{\mathbb{R}^n} |g_n(x)|^p dx \leq 1.$$

Thus $g \in L^p(\mathbb{R}^n)$ and $g(x) < \infty$ for almost every x , which implies the series $\sum_j (f_{k_{j+1}} - f_{k_j})(x)$ converges absolutely a.e.

Define (for those x where the series converges)

$$f(x) := f_{k_1}(x) + \sum_{j=1}^{\infty} (f_{k_{j+1}}(x) - f_{k_j}(x)),$$

and set $f(x) = 0$ where it does not converge. Then f is measurable and finite a.e., and

$$f(x) = \lim_{i \rightarrow \infty} f_{k_i}(x) \quad \text{a.e.}$$

To show $f_{k_j} \rightarrow f$ in L^p , define the *tail sum*

$$r_n(x) := \sum_{j=n}^{\infty} |f_{k_{j+1}}(x) - f_{k_j}(x)|.$$

Since $|f_{k_n}(x) - f(x)| \leq r_n(x)$ a.e., we have

$$\|f_{k_n} - f\|_{L^p} \leq \|r_n\|_{L^p} \leq \sum_{j=n}^{\infty} \|f_{k_{j+1}} - f_{k_j}\|_{L^p} \leq \sum_{j=n}^{\infty} 2^{-j} = 2^{-n+1} \rightarrow 0.$$

Therefore $f_{k_n} \rightarrow f$ in L^p . Since the original sequence (f_k) is Cauchy, the entire sequence converges to the same limit f in L^p . Hence $L^p(\mathbb{R}^n)$ is complete for $1 \leq p < \infty$.

Case $p = \infty$. If (f_k) is Cauchy in $L^\infty(\mathbb{R}^n)$, then for every $\varepsilon > 0$ there exists N such that $\|f_m - f_n\|_\infty < \varepsilon$ for all $m, n \geq N$, so there is a measurable set E where

$$|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_{L^\infty}$$

and the complement satisfies $\mu(E^c) = 0$. Then $(f_m(x))_{m=1}^\infty$ is a Cauchy sequence in $\mathbb{C}$ for $x \in E$, converging to some value, call it $f(x) \in \mathbb{C}$. For $x \in E^c$ we define $f(x) = 0$. Then f is measurable and

$$\lim_{n \rightarrow \infty} \|f - f_n\|_{L^\infty} = 0.$$

It only remains to verify that $f \in L^\infty(\mathbb{R}^n)$, but this is immediate from

$$|f(x)| \leq \|f_n\|_{L^\infty} + \|f - f_n\|_{L^\infty}.$$

Thus $f_k \rightarrow f$ in $L^\infty(\mathbb{R}^n)$, and $L^\infty(\mathbb{R}^n)$ is complete as well. $\square$

Remark 1.1.2. Since L^2 is a complete normed space equipped with an inner product that induces the norm, the space L^2 is a *Hilbert space*. For $p \neq 2$, the spaces L^p are complete normed vector spaces (Banach spaces), but they do not admit any natural inner product compatible with the norm $\|\cdot\|_{L^p}$.

1.2 Density of test functions

Definition 1.2.1 (Density in a metric space). Let (X, d) be a metric space and let $A \subset X$. We say that A is *dense in X* if for every $x \in X$ and every $\varepsilon > 0$ there exists $a \in A$ such that

$$d(x, a) < \varepsilon.$$

Equivalently, the closure of A is all of X : $\overline{A} = X$.

Example 1.2.2 (Density in normed spaces). Let $(X, \|\cdot\|)$ be a normed vector space and let $A \subset X$. We say that A is *dense in X* if for every $x \in X$ and every $\varepsilon > 0$ there exists $a \in A$ such that

$$\|x - a\| < \varepsilon.$$

Equivalently, the closure of A with respect to the norm topology is X .

Before moving forward, let us introduce some convenient notation. Let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ be a n -tuple of non-negative integers, called a *multi-index*. The magnitude of a multi-index is

$$|\alpha| = \alpha_1 + \dots + \alpha_n.$$

This allows us to write cumbersome partial derivatives in more compact way:

$$\partial^\alpha f = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n} f.$$

Similarly, for a vector $x \in \mathbb{R}^n$, we will denote

$$x^\alpha = x_1^{\alpha_1} \cdot x_2^{\alpha_2} \dots x_{n-1}^{\alpha_{n-1}} \cdot x_n^{\alpha_n}.$$

Let us denote the space of k times continuously differentiable functions by

$$C^k(\mathbb{R}^n) := \{f : \mathbb{R}^n \rightarrow \mathbb{C} \mid \partial^\alpha f \text{ is continuous } \forall |\alpha| \leq k\}$$

and the space of *smooth functions* by

$$C^\infty(\mathbb{R}^n) := \bigcap_{k=0}^{\infty} C^k(\mathbb{R}^n).$$

Let us also denote the *support* of $f : \mathbb{R}^n \rightarrow \mathbb{C}$ by

$$\text{supp}(f) := \overline{\{x \in \mathbb{R}^n \mid f(x) \neq 0\}}.$$

The space of smooth, compactly supported functions, is

$$C_c^\infty(\mathbb{R}^n) := \{f : \mathbb{R}^n \rightarrow \mathbb{C} \mid f \in C^\infty(\mathbb{R}^n), \text{supp}(f) \text{ is compact}\}.$$

These functions are also called *test functions*. It will be useful to know that L^p functions can be approximated by smooth and compactly supported functions.

Exercise 1.2.3. Show that

$$\varphi(x) = \begin{cases} e^{\frac{1}{|x|^2-1}+1}, & \text{if } |x| < 1, \\ 0, & \text{otherwise} \end{cases}$$

is smooth and supported on the ball $\overline{B_1(0)} = \{x \in \mathbb{R}^n \mid |x| \leq 1\}$.

Exercise 1.2.4. Let $\varphi \in C_c^\infty(\mathbb{R}^n)$. Show that $\varphi(x) = \varphi(0) + x \cdot \psi(x)$, where $\psi \in (C_c^\infty(\mathbb{R}^n))^n$.

The proofs of the following theorems require some measure-theoretic tools and will be omitted, but the theorems themselves illustrate useful approximation ideas that will reappear throughout the course.

Theorem 1.2.5. *If $X = \mathbb{R}^n$ with Lebesgue measure and $1 \leq p < \infty$, then*

$$C_c^\infty(\mathbb{R}^n) \text{ is dense in } L^p(\mathbb{R}^n).$$

Proof. See [1, Theorem 1.2.19 and Exercise 1.2.4]. □

Remark 1.2.6. In addition, in Theorem 1.2.5 it can be proved that for any measurable set $E \subset \mathbb{R}^n$ of finite measure there exists a sequence $(\varphi_n)_{n=1}^\infty \subset C_c^\infty(\mathbb{R}^n)$ such that $0 \leq \varphi_n \leq 1$ and $\varphi_n \rightarrow \mathbf{1}_E$ in L^p .

Remark 1.2.7. The case $p = \infty$ is different: C_c^∞ is not dense in L^∞ , but it is dense in L^∞ equipped with the weak-* topology.

Theorem 1.2.8. *If $(f_n)_{n=1}^\infty \subset L^p(\mathbb{R}^n)$ converges to $f \in L^p(\mathbb{R}^n)$, then there is a subsequence $(f_{n_k})_{k=1}^\infty$ such that $f_{n_k} \rightarrow f$ a.e.*

Proof. See [1, Theorem 1.1.11], [3, Theorem 3.12], or any book on measure theory. The intuition is that if $f_k \not\rightarrow f$ a.e., then there would be “bad sets” of positive measure, where $f_n \not\rightarrow f$. The exact argument uses Chebyshev’s inequality and Borel-Cantelli lemma to construct the a.e. convergent subsequence. □

If a function $f \in L^1(K)$ for any compact set $K \subset \mathbb{R}^n$, we call f *locally integrable* and denote $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. The following theorem is a useful application of density of test functions, giving a meaning to the concept *test function*, since one can test locally integrable functions by integrating them against smooth compactly supported functions.

Theorem 1.2.9. *Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. If $\int f\varphi dx = 0$ for all $\varphi \in C_c^\infty(\mathbb{R}^n)$, then $f = 0$ a.e.*

Sketch of the proof. Without loss of generality, suppose f is real-valued. Let $K \subset \mathbb{R}^n$ be compact and $E' = \{x \in \mathbb{R}^n \mid f(x) > 0\}$, and denote $E = E' \cap K$. Choose $(\varphi_n)_{n=1}^\infty \subset C_c^\infty(\mathbb{R}^n)$ such that $0 \leq \varphi_n \leq 1$ and $\varphi_n \rightarrow \mathbf{1}_E$ a.e. Suppose also that $\text{supp}(\varphi_n) \subset C$ for some bounded $C \supset K$. Since $|f(x)\varphi_n(x)| \leq |f(x)|\mathbf{1}_C$ and $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, then Lebesgue's theorem on dominated convergence gives

$$0 = \int f\varphi_n dx \rightarrow \int_E f dx.$$

If $\mu(E) = \mu(E' \cap K) > 0$, then $\int_E f dx > 0$, a contradiction. Therefore $\mu(E) = 0$. Since this holds for all compact K , then by subadditivity of the measure $\mu(E') = 0$. Applying same argument to $-f$ gives $f = 0$. $\square$

Exercise 1.2.10. Find an example of a function $f \in L^\infty(\mathbb{R}^n)$ which cannot be approximated by smooth compactly supported functions in the L^∞ norm.

The density of smooth functions in L^p allows an alternative definition of the Lebesgue spaces: one may define

$L^p(\mathbb{R}^n)$ as the completion of $C_c^\infty(\mathbb{R}^n)$ with respect to the norm $\|\cdot\|_{L^p}$.

This viewpoint is conceptually elegant but somewhat abstract, since it is not immediately clear that the elements of the completion can be identified with actual functions (rather than limits of Cauchy sequences). The measure-theoretic definition, in contrast, makes this identification explicit by working with equivalence classes of measurable functions that agree almost everywhere.

1.3 Convolution and Young's inequality

For $f, g \in L^1(\mathbb{R}^n)$, we define the *convolution*

$$(f * g)(x) := \int_{\mathbb{R}^n} f(x-y)g(y) dy.$$

This is a commutative and associative operation on $L^1(\mathbb{R}^n)$.

Theorem 1.3.1 (Young's convolution inequality). *Let $1 \leq p, q, r \leq \infty$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}.$$

*If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, then $f * g \in L^r(\mathbb{R}^n)$ and*

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Proof. We prove the stated estimate in the case $q = 1$ and $r = p$, i.e.

$$\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1},$$

for $1 \leq p \leq \infty$. The case $p = \infty$ is immediate since

$$|f * g(x)| \leq \int_{\mathbb{R}^n} |f(x - y)| |g(y)| dy \leq \|f\|_{L^\infty} \|g\|_{L^1}.$$

Assume now $1 \leq p < \infty$. By Fubini and Hölder's inequality we get

$$\begin{aligned} \int_{\mathbb{R}^n} |(f * g)(x)|^p dx &= \int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} f(x - y) g(y) dy \right|^p dx \\ &\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x - y)| |g(y)| dy \right)^p dx \\ &\leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x - y)|^p |g(y)| dy \right) \left(\int_{\mathbb{R}^n} |g(y)| dy \right)^{p/p'} dx \\ &= \|g\|_{L^1}^{p/p'} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |f(x - y)|^p |g(y)| dy dx \\ &= \|g\|_{L^1}^{p/p'} \int_{\mathbb{R}^n} |g(y)| \int_{\mathbb{R}^n} |f(x - y)|^p dx dy \\ &= \|g\|_{L^1}^{p/p' + 1} \|f\|_{L^p}^p = \|f\|_{L^p}^p \|g\|_{L^1}^p. \end{aligned}$$

Taking p th roots gives

$$\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1}.$$

□

Exercise 1.3.2. Prove Theorem 1.3.1.

Chapter 2

Schwartz spaces and the Fourier transform

The Fourier transform is best behaved on a class of functions which are both *smooth* and *rapidly decaying*. This leads us to the Schwartz space.

Definition 2.0.1 (Schwartz space). The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of all smooth functions $f \in C^\infty(\mathbb{R}^n)$ such that for every pair of multi-indices $\alpha, \beta \in \mathbb{N}^n$, the seminorm

$$|f|_{\alpha, \beta} := \sup_{x \in \mathbb{R}^n} |x^\alpha \partial^\beta f(x)|$$

is finite. In other words, every derivative of f decays faster than any polynomial.

The Schwartz spaces are named after the French mathematician Laurent-Moïse Schwartz (1915—2002) (not to be confused with Hermann Schwarz (1843–1921)).

Exercise 2.0.2. Verify that $|f|_{\alpha, \beta}$ is a seminorm on $\mathcal{S}(\mathbb{R}^n)$, when $\alpha \geq 0$ and $\beta > 0$.

Example 2.0.3. The space $\mathcal{S}(\mathbb{R}^n)$ is not a normed space (nor is it normable), but it is a metric space with metric defined by

$$d(f, g) := \sum_{\alpha, \beta \geq 0} 2^{-|\alpha| - |\beta|} \frac{|f - g|_{\alpha, \beta}}{1 + |f - g|_{\alpha, \beta}}.$$

Thus, the Schwartz space is a Fréchet space, but it is not a Banach space.

Theorem 2.0.4. *The Schwartz space is a complete metric space.*

Exercise 2.0.5. Prove that the Gaussian $f(x) = e^{-|x|^2}$ belongs to $\mathcal{S}(\mathbb{R}^n)$. This function plays a central role as an extremal function in Fourier analysis.

2.1 The Fourier transform

Definition 2.1.1 (Fourier transform). For $f \in \mathcal{S}(\mathbb{R}^n)$, the Fourier transform is defined by

$$\widehat{f}(\xi) := \mathcal{F}(f)(\xi) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx,$$

The Fourier transform is well-defined as a bounded linear operator $\mathcal{F} : L^1(\mathbb{R}) \rightarrow L^\infty(\mathbb{R})$.

Definition 2.1.2. Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be any function, $h \in \mathbb{R}^n$ and $\lambda \in \mathbb{R} \setminus \{0\}$. Define *translation* τ_h and *dilation* σ_λ by

$$(\tau_h f)(x) = f(x - h), \quad (\sigma_\lambda f)(x) = f(\lambda x).$$

Theorem 2.1.3. Let $f \in \mathcal{S}(\mathbb{R}^n)$, $h \in \mathbb{R}^n$ and $\lambda \in \mathbb{R} \setminus \{0\}$. Then the following properties hold

- (a) $\widehat{\tau_h f}(\xi) = e^{-ih \cdot \xi} \widehat{f}(\xi)$ (translation)
- (b) $\widehat{\sigma_\lambda f}(\xi) = \frac{1}{|\lambda|^n} \sigma_{\frac{1}{\lambda}} \widehat{f}(\xi)$ (scaling/dilation)
- (c) $\widehat{\partial^\alpha f}(\xi) = (i\xi)^\alpha \widehat{f}(\xi)$ (differentiation)
- (d) $\widehat{x^\alpha f}(\xi) = i^{|\alpha|} \partial_\xi^\alpha \widehat{f}(\xi)$ (multiplication).

Proof. Exercise. □

Oftentimes it is convenient to use the notation

$$D^\alpha f = (-i\partial)^\alpha f.$$

It follows immediately from Theorem 2.1.3 (c) that if $f \in C^k(\mathbb{R}^n)$ and $D^\alpha f \in L^1(\mathbb{R}^n)$ for all $|\alpha| \leq k$, then, for large enough $|\xi|$

$$|\widehat{f}(\xi)| \leq C|\xi|^{-k} \quad \text{that is} \quad |\widehat{f}(\xi)| = O(|\xi|^{-k}).$$

Remark 2.1.4. Later in Chapter 5 we will define the so-called *Fourier multiplier operators* $m(D)$, where an operation on the function itself becomes multiplication by a function in the frequency domain.

Theorem 2.1.5. The Fourier transform maps Schwartz functions to Schwartz functions, that is,

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n).$$

Proof. Let $\varphi \in \mathcal{S}(\mathbb{R}^n)$. We need to show that $\widehat{\varphi} = \mathcal{F}(\varphi)$ is smooth and that all derivatives of $\widehat{\varphi}$ decay faster than any polynomial.

Differentiation under the integral sign gives, for any multi-index α ,

$$\partial_\xi^\alpha \widehat{\varphi}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \varphi(x) \partial_\xi^\alpha (e^{-ix \cdot \xi}) dx = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \varphi(x) (-ix)^\alpha e^{-ix \cdot \xi} dx.$$

Since $x^\alpha \varphi(x) \in L^1(\mathbb{R}^n)$ for all α , the integral converges absolutely. Hence $\widehat{\varphi}$ is smooth.

Multiplication by x^β in the integral corresponds to differentiation in ξ :

$$\xi^\beta \widehat{\varphi}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} \varphi(x) \xi^\beta e^{-ix \cdot \xi} dx.$$

More usefully, integration by parts gives

$$\xi^\beta \widehat{\varphi}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} (-i\partial_x)^\beta \varphi(x) e^{-ix \cdot \xi} dx,$$

so

$$|\xi^\beta \partial_\xi^\alpha \widehat{\varphi}(\xi)| \leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} |x^\alpha D^\beta \varphi(x)| dx < \infty.$$

Since this holds for all multi-indices α, β , all derivatives of $\widehat{\varphi}$ decay faster than any polynomial. Therefore $\widehat{\varphi} \in \mathcal{S}(\mathbb{R}^n)$. Hence $\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$. $\square$

Exercise 2.1.6. Show that

$$\mathcal{F}(e^{-\frac{1}{2}|x|^2})(\xi) = e^{-\frac{1}{2}|\xi|^2}.$$

Therefore the function $e^{-\frac{1}{2}|x|^2}$ acts like a unit for the Fourier transform.

Theorem 2.1.7. Let $f, g \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\widehat{f * g} = (2\pi)^{\frac{n}{2}} \widehat{f} \widehat{g}.$$

2.2 Fourier inversion and Plancherel theorem

We begin with a rather technical, but useful, lemma:

Lemma 2.2.1. Let $f \in \mathcal{S}(\mathbb{R}^n)$ and

$$\phi_\varepsilon(x) = (2\pi\varepsilon)^{-n/2} e^{-|x|^2/(2\varepsilon)}.$$

Then

$$f * \phi_\varepsilon \longrightarrow f$$

uniformly, as $\varepsilon \rightarrow 0^+$.

Proof. Let us denote $f_\varepsilon := f * \phi_\varepsilon$. Since $\int_{\mathbb{R}^n} \phi_\varepsilon(y) dy = 1$, we have

$$f_\varepsilon(x) - f(x) = \int_{\mathbb{R}^n} (f(x-y) - f(x)) \phi_\varepsilon(y) dy. \quad (2.1)$$

Let $\delta > 0$. Since f is uniformly continuous, there exists $\eta > 0$ such that

$$|f(x-y) - f(x)| < \frac{\delta}{2} \quad \text{whenever } |y| < \eta$$

for any $x \in \mathbb{R}^n$. Split the integral (2.1) as

$$\int_{|y| < \eta} + \int_{|y| \geq \eta}.$$

For the first term,

$$\int_{|y| < \eta} |f(x-y) - f(x)| \phi_\varepsilon(y) dy \leq \frac{\delta}{2} \int_{\mathbb{R}^n} \phi_\varepsilon(y) dy = \frac{\delta}{2}.$$

For the second term, using boundedness of f and rapid decay of the Gaussian,

$$\begin{aligned} \int_{|y| \geq \eta} |f(x-y) - f(x)| \phi_\varepsilon(y) dy &\leq 2\|f\|_{L^\infty(\mathbb{R}^n)} \int_{|y| \geq \eta} \phi_\varepsilon(y) dy \\ &= 2\|f\|_{L^\infty(\mathbb{R}^n)} \int_{|y| \geq \eta} (2\pi\varepsilon)^{-n/2} e^{-|y|^2/(2\varepsilon)} dy \\ &= 2\|f\|_{L^\infty(\mathbb{R}^n)} \int_{|z| \geq \frac{\eta}{\sqrt{\varepsilon}}} (2\pi)^{-n/2} e^{-|z|^2/2} dz < \frac{\delta}{2} \end{aligned}$$

when $\varepsilon > 0$ is sufficiently small. In the second to last integration we changed the variable of integration as $z = y/\sqrt{\varepsilon}$. We have shown that there is $\varepsilon > 0$ so that

$$|f_\varepsilon(x) - f(x)| < \delta,$$

for any $x \in \mathbb{R}^n$. Since $\delta > 0$ was arbitrary, this shows that $f_\varepsilon \rightarrow f$ uniformly. $\square$

Now we can define the *inverse Fourier transform*.

Theorem 2.2.2 (Fourier inversion). *If $f \in \mathcal{S}(\mathbb{R}^n)$, then*

$$f(\xi) = \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} \widehat{f}(\xi) e^{ix \cdot \xi} d\xi =: \mathcal{F}^* \mathcal{F}(f)(x).$$

Proof. The idea is to reconstruct f from its Fourier transform using the fact that the Gaussian is its own Fourier transform.

Let $\phi_\varepsilon(x) = (2\pi\varepsilon)^{-n/2} e^{-|x|^2/(2\varepsilon)}$ be a Gaussian. By Exercise 2.1.6 and a change of variables, its Fourier transform is

$$\widehat{\phi_\varepsilon}(\xi) = (2\pi)^{-\frac{n}{2}} e^{-\varepsilon|\xi|^2/2}.$$

and, conversely,

$$\mathcal{F}(e^{-\varepsilon|\xi|^2/2})(x) = (2\pi)^{\frac{n}{2}} \phi_\varepsilon(x).$$

Now by Lemma 2.2.1

$$\begin{aligned}
f(0) &= \lim_{\varepsilon \rightarrow 0^+} (f * \phi_\varepsilon)(0) \\
&= \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(x) \phi_\varepsilon(x) dx \\
&= \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(x) (2\pi)^{-\frac{n}{2}} \mathcal{F}(e^{-\varepsilon|\xi|^2/2})(x) dx \\
&= (2\pi)^{-\frac{n}{2}} \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} f(x) (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} e^{-\varepsilon|\xi|^2/2} d\xi dx \\
&= (2\pi)^{-\frac{n}{2}} \lim_{\varepsilon \rightarrow 0^+} \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx e^{-\varepsilon|\xi|^2/2} d\xi \\
&= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx d\xi \\
&= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \mathcal{F}(f)(\xi) e^{i0 \cdot \xi} d\xi
\end{aligned}$$

where we used Fubini's theorem to interchange the order of integration, and Lebesgue's theorem on dominated convergence to take limit inside the integral (and note that $e^{i0 \cdot \xi} = 1$ in the last step). We have obtained

$$f(0) = \mathcal{F}^* \mathcal{F}(f)(0).$$

The proof is finished by performing a translation and using Theorem 2.1.3:

$$\begin{aligned}
f(x) &= \tau_{-x}(f)(0) = \mathcal{F}^* \mathcal{F}(\tau_{-x}f)(0) \\
&= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} e^{ix \cdot \xi} \mathcal{F}(f)(x) dx = \mathcal{F}^* \mathcal{F}(f)(x),
\end{aligned}$$

which is what we wanted. $\square$

Remark 2.2.3. The above theorem allows us to define the *inverse Fourier transform*

$$\mathcal{F}^{-1}(f)(x) := \frac{1}{(2\pi)^{\frac{n}{2}}} \int_{\mathbb{R}^n} f(\xi) e^{ix \cdot \xi} d\xi.$$

Corollary 2.2.4. *The Fourier transform is an automorphism of $\mathcal{S}(\mathbb{R}^n)$, that is,*

$$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$$

is a continuous map with continuous inverse, and

$$\mathcal{F} \mathcal{F}^{-1} = \mathcal{F}^{-1} \mathcal{F} = I.$$

Proof. We already know from previous theorems that $\mathcal{F}$ is bijective with inverse $\mathcal{F}^{-1}$. Continuity follows from the fact that differentiation and multiplication by polynomials in x correspond to similar operations in Fourier space, which are continuous operations in the Schwartz topology. Likewise, the inverse map $\mathcal{F}^{-1}$

is continuous for the same reason (note also that $\widehat{f}(-\xi) = \mathcal{F}^{-1}(f)(\xi)$, so any property proven for $\mathcal{F}$ directly translates to $\mathcal{F}^{-1}$).

Hence $\mathcal{F}$ is a continuous automorphism of $\mathcal{S}(\mathbb{R}^n)$, as all bounded sets (in the topology of $\mathcal{S}$) map to bounded sets. $\square$

An important property of the Fourier transform is that it acts as a *unitary operator* on the space $L^2(\mathbb{R}^n)$. At the moment the Fourier transform is only defined in the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ (or in $L^1(\mathbb{R}^n)$, where the integrals are also well-defined), but later on we will extend the Fourier transform to $L^p(\mathbb{R}^n)$ for $1 \leq p \leq 2$.

Theorem 2.2.5 (Parseval's equality). *Let $f \in \mathcal{S}(\mathbb{R}^n)$. Then*

$$\|f\|_{L^2(\mathbb{R}^n)} = \|\widehat{f}\|_{L^2(\mathbb{R}^n)}.$$

Proof. Consider first the adjoint operator of $\mathcal{F}$ in the L^2 inner product. Let $f, g \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\langle \mathcal{F}f, g \rangle_{L^2} = \int_{\mathbb{R}^n} \widehat{f}(x) \overline{g(\xi)} d\xi = (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} f(x) dx \overline{g(\xi)} d\xi \quad (2.2)$$

$$= (2\pi)^{-\frac{n}{2}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \overline{e^{ix \cdot \xi} g(\xi)} d\xi f(x) dx =: \langle f, \mathcal{F}^* g \rangle_{L^2}, \quad (2.3)$$

where we used Fubini's theorem to change the order of integration, and where we denote $\mathcal{F}^* g(x) = \mathcal{F}g(-x)$.

Now we can observe that $\mathcal{F}^* = \mathcal{F}^{-1}$, and therefore

$$\|\mathcal{F}f\|_{L^2}^2 = \langle \mathcal{F}f, \mathcal{F}f \rangle_{L^2} = \langle f, \mathcal{F}^* \mathcal{F}f \rangle_{L^2} = \langle f, \mathcal{F}^{-1} \mathcal{F}f \rangle_{L^2} = \langle f, f \rangle_{L^2} = \|f\|_{L^2}^2.$$

This equality of norms is sometimes also called *Parseval's equality*. $\square$

Chapter 3

Fourier transform on L^p

In this chapter, we collect the fundamental L^p -mapping properties of the Fourier transform and the related convolution inequalities that are routinely used in analysis and PDEs.

Theorem 3.0.1 (Riemann–Lebesgue lemma). *If $f \in L^1(\mathbb{R}^n)$, then $\widehat{f}$ is bounded, continuous, and*

$$\lim_{|\xi| \rightarrow \infty} \widehat{f}(\xi) = 0.$$

In other words $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow C_0(\mathbb{R}^n)$.

Proof. Directly

$$|\widehat{f}(\xi)| \leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} |f(x)| dx = (2\pi)^{-n/2} \|f\|_{L^1} < \infty,$$

so $\widehat{f}$ is bounded.

If $\xi, h \in \mathbb{R}^n$, then

$$\begin{aligned} |\widehat{f}(\xi + h) - \widehat{f}(\xi)| &\leq (2\pi)^{-n/2} \int_{\mathbb{R}^n} |f(x)| |e^{-ix \cdot h} - 1| dx \\ &\leq \int_{|x||h| \leq \sqrt{|h|}} |f(x)| |x||h| dx + \int_{|x||h| \geq \sqrt{|h|}} |f(x)| dx \\ &\leq \sqrt{|h|} \|f\|_{L^1(\mathbb{R}^n)} + 2 \int_{|x| \geq 1/\sqrt{|h|}} |f(x)| dx \rightarrow 0 \end{aligned}$$

as $h \rightarrow 0$, because $f \in L^1(\mathbb{R}^n)$ and $|e^{-ix \cdot h} - 1| \leq |x||h|$, when $|x||h| \leq 1$. This shows that $\widehat{f}$ is (uniformly) continuous.

We can obtain more, namely that $\widehat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$. To show this,

consider

$$\begin{aligned}
2\widehat{f}(\xi) &= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx - (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-i(x - \pi\xi/|\xi|^2) \cdot \xi} dx \\
&= (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix \cdot \xi} dx - (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(y + \pi\xi/|\xi|^2) e^{-iy \cdot \xi} dy \\
&= -(2\pi)^{-n/2} \int_{\mathbb{R}^n} (f(x + \pi\xi/|\xi|^2) - f(x)) e^{-ix \cdot \xi} dx.
\end{aligned}$$

But now, due to Lebesgue's theorem on dominated convergence, we see that

$$2|\widehat{f}(\xi)| \leq C \int_{\mathbb{R}^n} |f(x + \pi\xi/|\xi|^2) - f(x)| dx \rightarrow 0$$

as $|\xi| \rightarrow \infty$. □

Remark 3.0.2. The Riemann–Lebesgue lemma shows that Fourier transforms of L^1 functions are always continuous and vanish at infinity, even if the original function is highly irregular or discontinuous. This is another indication that the Fourier transform interchanges integrability with smoothness and decay.

Theorem 3.0.3 (Plancherel). *The Fourier transform extends uniquely to a unitary operator $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$, satisfying*

$$\|f\|_{L^2} = \|\widehat{f}\|_{L^2}, \quad \langle f, g \rangle_{L^2} = \langle \widehat{f}, \widehat{g} \rangle_{L^2}.$$

Proof. Recall that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ and on $\mathcal{S}(\mathbb{R}^n)$ the Fourier transform is well-defined and maps into $L^2(\mathbb{R}^n)$ (any Schwartz function belongs to $L^2(\mathbb{R}^n)$). We already proved Parseval's equality for functions $f \in \mathcal{S}(\mathbb{R}^n)$ in Theorem 2.2.5.

Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ and $\mathcal{F}$ is an isometry, it extends uniquely to a bounded linear map $\mathcal{F}_{\text{ext}} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ preserving the norm. More precisely, let $f \in L^2(\mathbb{R}^n)$ and take a sequence $(f_k)_{k=1}^\infty \subset \mathcal{S}(\mathbb{R}^n)$ with $f_k \rightarrow f$ in $L^2(\mathbb{R}^n)$ as $k \rightarrow \infty$. This means that $(f_k)_{k=1}^\infty$ is Cauchy in L^2 . Due to Parseval's equality we have

$$\|\mathcal{F}(f_k) - \mathcal{F}(f_l)\|_{L^2} = \|\mathcal{F}(f_k - f_l)\|_{L^2} = \|f_k - f_l\|_{L^2} \rightarrow 0$$

as $k, l \rightarrow \infty$. Therefore, $(\mathcal{F}(f_k))_{k=1}^\infty$ is a Cauchy sequence in $L^2(\mathbb{R}^n)$ with a limit

$$\mathcal{F}_{\text{ext}}(f) := \lim_{k \rightarrow \infty} \mathcal{F}(f_k) \in L^2(\mathbb{R}^n).$$

From now on we will simply use the notation $\mathcal{F} = \mathcal{F}_{\text{ext}}$.

Finally, the image of the isometry $\mathcal{F}$ is a closed subspace of $L^2(\mathbb{R}^n)$. Since the image contains the dense subspace $\mathcal{S}(\mathbb{R}^n)$, the image must be all of $L^2(\mathbb{R}^n)$. Therefore, $\mathcal{F}$ is a unitary operator on $L^2(\mathbb{R}^n)$ and

$$\langle f, g \rangle_{L^2} = \langle \widehat{f}, \widehat{g} \rangle_{L^2}$$

holds. □

Remark 3.0.4. Plancherel's theorem is the foundation for Fourier analysis in L^2 spaces, allowing one to treat Fourier transforms as unitary operators on Hilbert spaces. It also justifies defining the Fourier transform of tempered distributions by duality.

We next extend the Fourier transform from L^1 and L^2 to any L^p with $1 < p < 2$. For this, we require

Theorem 3.0.5 (Riesz–Thorin). *Let (X, μ) and (Y, ν) be measure spaces. Suppose $T : L^{p_0} + L^{p_1} \rightarrow L^{q_0} + L^{q_1}$ is a linear operator that boundedly maps $L^{p_0} \rightarrow L^{q_0}$ and $L^{p_1} \rightarrow L^{q_1}$ with the following end-point bounds:*

$$\|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}}, \quad \|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}},$$

for some $1 \leq p_0, p_1, q_0, q_1 \leq \infty$. For $0 < \theta < 1$ define

$$\frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}, \quad \frac{1}{q} = \frac{1-\theta}{q_0} + \frac{\theta}{q_1}.$$

Then T extends to a bounded operator $L^p(X) \rightarrow L^q(Y)$ and

$$\|T\|_{L^p \rightarrow L^q} \leq M_0^{1-\theta} M_1^\theta.$$

Remark 3.0.6. The proof uses complex-analytic interpolation (Hadamard three-lines lemma) and will not be given here. Riesz–Thorin is the standard tool to interpolate linear operators between Lebesgue spaces; we will use it to deduce the Hausdorff–Young inequality.

Theorem 3.0.7 (Hausdorff–Young). *Let $1 \leq p \leq 2$ and let p' denote the conjugate exponent, $1/p + 1/p' = 1$. For every $f \in L^p(\mathbb{R}^n)$ the Fourier transform $\widehat{f}$ belongs to $L^{p'}(\mathbb{R}^n)$ and*

$$\|\widehat{f}\|_{L^{p'}(\mathbb{R}^n)} \leq (2\pi)^{-n(\frac{1}{p}-\frac{1}{2})} \|f\|_{L^p(\mathbb{R}^n)}.$$

Equivalently the Fourier transform extends to a bounded linear map $\mathcal{F} : L^p \rightarrow L^{p'}$.

Sketch of proof via Riesz–Thorin. Directly from definition we have $\mathcal{F} : L^1(\mathbb{R}^n) \rightarrow L^\infty(\mathbb{R}^n)$ and from Plancherel's theorem $\mathcal{F} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$, so that $\mathcal{F} : L^1 + L^2 \rightarrow L^2 + L^\infty$ with the following end-point estimates

- $L^1 \rightarrow L^\infty$: $\|\widehat{f}\|_\infty \leq \frac{1}{(2\pi)^{\frac{n}{2}}} \|f\|_1$ (from the definition);
- $L^2 \rightarrow L^2$: $\|\widehat{f}\|_2 = \|f\|_2$ (Plancherel).

Now apply Riesz–Thorin with $(p_0, q_0) = (1, \infty)$ and $(p_1, q_1) = (2, 2)$. Then

$$\frac{1}{p} = \frac{1-\theta}{1} + \frac{\theta}{2} = 1 - \frac{\theta}{2}, \quad \frac{1}{q} = \frac{1-\theta}{\infty} + \frac{\theta}{2} = \frac{\theta}{2}.$$

Hence $q = p'$ and the interpolation gives $\|\widehat{f}\|_{p'} \leq (2\pi)^{-n(\frac{1}{p}-\frac{1}{2})} \|f\|_p$ for all $1 \leq p \leq 2$. $\square$

Remark 3.0.8. The result is optimal: the Fourier transform does not extend to a bounded linear mapping $L^p \rightarrow L^{p'}$ when $p > 2$. Equality of norms occurs at $p = 2$. For general p the Fourier transform is generally interpreted in the sense of tempered distributions which we will study later in Chapter 4.

Remark 3.0.9 (Truncated Fourier transform). For $f \in L^2(\mathbb{R}^n)$ and $R > 0$, one can define the truncated Fourier transform

$$\mathcal{F}_R(f)(\xi) = \frac{1}{(2\pi)^{n/2}} \int_{|x| < R} f(x) e^{-ix \cdot \xi} dx.$$

Plancherel's theorem ensures that $\mathcal{F}_R(f) \rightarrow \hat{f}$ in L^2 norm as $R \rightarrow \infty$.

However, for $p \neq 2$, this convergence need not be true. In fact, for $p \neq 2$ the frequency truncated Fourier transforms are not uniformly bounded as operators $\mathcal{F}_R : L^p \rightarrow L^p$ (see counterexamples of C. Fefferman¹ that use special Besicovitch needle sets.). This highlights the special role of L^2 among all L^p spaces.

¹C. Fefferman, *The multiplier problem for the ball*, Annals of Mathematics, 94 (2): 330–336, 1972, doi:10.2307/1970864, and the book Grafakos, Classical Fourier Analysis [1, Theorem 8.20]

Chapter 4

Tempered distributions

How do derivatives make sense for non-smooth functions? What about the Fourier transform of something that is not integrable, or even not a function at all? The framework of *distributions* (or *generalized functions*) provides a rigorous way to extend differentiation and the Fourier transform far beyond classical L^p functions.

Definition 4.0.1 (Tempered distributions). A *tempered distribution* is a continuous linear functional

$$T : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathbb{C}, \quad \varphi \mapsto \langle T, \varphi \rangle = T(\varphi),$$

with respect to the Fréchet topology of $\mathcal{S}(\mathbb{R}^n)$ induced by the family of seminorms $|\cdot|_{\alpha,\beta}$ as in Example 2.0.3.

The space of all tempered distributions is denoted by $\mathcal{S}'(\mathbb{R}^n)$, the continuous dual of $\mathcal{S}(\mathbb{R}^n)$. When $T \in \mathcal{S}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$, we denote their duality pairing by

$$\langle T, \varphi \rangle := T(\varphi).$$

It is definitional that a distribution is uniquely determined by its action on test functions: If $\langle T, \varphi \rangle = 0$ for all $\varphi \in \mathcal{S}(\mathbb{R}^n)$, then by definition $T(\varphi) = 0$ for all functions $\varphi \in \mathcal{S}(\mathbb{R}^n)$, so $T = 0$.

Every $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ of at most polynomial growth defines a tempered distribution by

$$\langle f, \varphi \rangle := \int_{\mathbb{R}^n} f(x)\varphi(x) dx.$$

One should be slightly careful here: the distributional pairing is a duality pairing and does not involve complex conjugation, unlike the usual L^2 inner product, which conjugates in the second entry. Thus, $\mathcal{S}'(\mathbb{R}^n)$ extends the notion of “function” but allows also non-function objects such as the Dirac delta.

The *Dirac delta distribution* δ is defined by

$$\langle \delta, \varphi \rangle = \varphi(0), \quad \varphi \in \mathcal{S}(\mathbb{R}^n).$$

It acts as a unit mass concentrated at the origin. For any test function φ , the notion $\int \delta(x)\varphi(x) dx = \varphi(0)$ is understood in this distributional sense, and this informal notation is common in the literature.

Consider the function $x \mapsto 1/x$ on $\mathbb{R} \setminus \{0\}$. Its “principal value” defines a tempered distribution p.v. $\frac{1}{x}$ by

$$\langle \text{p.v. } \frac{1}{x}, \varphi \rangle := \lim_{\varepsilon \rightarrow 0^+} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

This distribution appears naturally when computing the Fourier transform of the sign function, or in general, of odd kernels.

4.1 Basic operations on tempered distributions

If $T \in \mathcal{S}'(\mathbb{R}^n)$ and $\psi \in C^\infty(\mathbb{R}^n)$ grows at most polynomially, we define

$$\langle \psi T, \varphi \rangle := \langle T, \psi \varphi \rangle.$$

This extends the usual product $f\psi$ when $T = f$ is an ordinary function.

Derivatives of a distribution are defined by transferring differentiation onto the test function:

$$\langle \partial^\alpha T, \varphi \rangle := (-1)^{|\alpha|} \langle T, \partial^\alpha \varphi \rangle.$$

This guarantees that the operation is well-defined and coincides with the classical derivative whenever $T = f \in \mathcal{S}(\mathbb{R}^n)$. The definition is motivated by the classical integration by parts formula

$$\int_{\mathbb{R}^n} \partial^\alpha f(x) \varphi(x) dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} f(x) \partial^\alpha \varphi(x) dx.$$

Exercise 4.1.1. Prove that the distributional derivative of the Heaviside function $H(x) = \mathbf{1}_{(0, \infty)}(x)$ is the delta:

$$H'(x) = \delta(x).$$

Exercise 4.1.2. Show that the derivatives of the delta distribution act on functions $\varphi \in \mathcal{S}(\mathbb{R}^n)$ as

$$\langle \partial^\alpha \delta, \varphi \rangle = (-1)^{|\alpha|} \partial^\alpha \varphi(0).$$

The Fourier transform extends continuously to $\mathcal{S}'(\mathbb{R}^n)$ by duality:

$$\langle \widehat{T}, \varphi \rangle := \langle T, \widehat{\varphi} \rangle, \quad T \in \mathcal{S}'(\mathbb{R}^n), \quad \varphi \in \mathcal{S}(\mathbb{R}^n),$$

where $\widehat{\varphi}$ denotes the classical Fourier transform of the test function. The restriction to tempered distributions (rather than all distributions arising from a narrower class of test functions) is convenient because the Fourier transform maps $\mathcal{S}'$ onto itself, keeping the theory symmetric.

This definition ensures the fundamental identities

$$\widehat{\partial^\alpha T} = (i\xi)^\alpha \widehat{T}, \quad \widehat{x^\alpha T} = i^{|\alpha|} \partial_\xi^\alpha \widehat{T},$$

which generalize the usual differentiation and multiplication rules. Let us verify the first equality, and let $T \in \mathcal{S}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\begin{aligned} \langle \mathcal{F}(\partial^\alpha T), \varphi \rangle &= \langle \partial^\alpha T, \widehat{\varphi} \rangle \\ &= (-1)^{|\alpha|} \langle T, \partial^\alpha \widehat{\varphi} \rangle \\ &= (-1)^{|\alpha|} \langle T, \mathcal{F}((-ix)^\alpha \varphi) \rangle \\ &= \langle \widehat{T}, (i\xi)^\alpha \varphi \rangle \\ &= \langle (i\xi)^\alpha \widehat{T}, \varphi \rangle. \end{aligned}$$

Exercise 4.1.3. Prove that the Fourier transform of δ is the constant $(2\pi)^{-n/2}$, while

$$\widehat{1} = (2\pi)^{n/2} \delta,$$

so the transform exchanges localization and spread: the constant function transforms into a point mass, and vice versa.

Chapter 5

Sobolev spaces

5.1 Weak derivatives

Classical derivatives are defined pointwise, and require the function to be differentiable in the usual sense. However, many functions arising in analysis and PDEs are not differentiable everywhere, yet they behave as if they had derivatives in an “averaged” sense. This motivates the concept of *weak derivatives*.

Definition 5.1.1 (Weak derivative). Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and let $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ be a multi-index. A function $g \in L^1_{\text{loc}}(\mathbb{R}^n)$ is called the *weak derivative* $D^\alpha f$ of f if

$$\int_{\mathbb{R}^n} f(x) D^\alpha \varphi(x) dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} g(x) \varphi(x) dx$$

for all test functions $\varphi \in C_c^\infty(\mathbb{R}^n)$.

The idea of a weak derivative is to extend differentiation to functions that are not differentiable in the classical sense, by *requiring that the integration by parts identity continues to hold*. In other words, weak derivatives are defined precisely so that one can still integrate by parts.

Weak derivatives are unique up to sets of measure zero, see Theorem 1.2.9 and [2, Section 5.2]. If a function defines a tempered distribution and has a weak derivative, the weak derivative coincides with the distributional derivative. However, a distributional derivative of a function does not need to be representable by a function $g \in L^1_{\text{loc}}$ and hence need not be a weak derivative.

Theorem 5.1.2. *Let $f \in L^1_{\text{loc}}(\mathbb{R})$ be weakly differentiable. If $f' = 0$ then f is constant almost everywhere.*

Proof. We need to show that if

$$\int_{-\infty}^{\infty} f(x) \varphi'(x) dx = 0 \quad \text{for all } \varphi \in C_c^\infty(\mathbb{R}),$$

then $f = C$ for some $C \in \mathbb{R}$ a.e. So let $\varphi \in C_c^\infty(\mathbb{R})$. Note now that the function $\psi = \varphi'$ belongs to $C_c^\infty(\mathbb{R})$ and satisfies

$$\int_{-\infty}^{\infty} \psi(x) dx = 0.$$

So the problem reduces to showing that if

$$\int_{-\infty}^{\infty} f(x)\psi(x)dx = 0 \quad \text{for all } \psi \in C_c^\infty(\mathbb{R}), \quad \int_{-\infty}^{\infty} \psi(x)dx = 0$$

then f is constant a.e.

We show that f is *translation invariant*: for any $h \in \mathbb{R}$, $f(x+h) = f(x)$ for almost all $x \in \mathbb{R}$. So let $h \in \mathbb{R}$ and $\eta \in C_c^\infty(\mathbb{R})$, and define

$$\psi_h(x) = \eta(x-h) - \eta(x).$$

Now $\psi_h \in C_c^\infty(\mathbb{R})$ and

$$\int_{-\infty}^{\infty} \psi_h(x)dx = \int_{-\infty}^{\infty} \eta(x-h)dx - \int_{-\infty}^{\infty} \eta(x)dx = 0.$$

Testing f against ψ_h we get

$$\int_{-\infty}^{\infty} f(x)(\eta(x-h) - \eta(x))dx = 0.$$

Therefore (after changes of variables)

$$\int_{-\infty}^{\infty} (f(x+h) - f(x))\eta(x)dx = 0 \quad \text{for all } \eta \in C_c^\infty(\mathbb{R}).$$

By Theorem 1.2.9 $f(x+h) = f(x)$ a.e. for all $h \in \mathbb{R}$. To finish the proof, pick any Lebesgue point $x_0 \in \mathbb{R}$ of f . Then $f(x) = f(x_0 + (x - x_0)) = f(x_0)$ a.e. and f must be constant. $\square$

Corollary 5.1.3. *Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. If $\nabla f = 0$ weakly, then f is constant almost everywhere.*

Remark 5.1.4. Note that weak derivatives are *not* the same thing as derivatives almost everywhere. For example, the Heaviside function H does not have a weak derivative, even though $H' = 0$ almost everywhere.

Exercise 5.1.5. Prove that the distributional derivative $H' = \delta$ and that $\delta \notin L^1_{\text{loc}}(\mathbb{R})$.

Example 5.1.6. The function

$$f(x) = |x|, \quad x \in \mathbb{R},$$

is not differentiable at $x = 0$. However, for any $\varphi \in C_c^\infty(\mathbb{R})$ we have

$$\int_{\mathbb{R}} |x|\varphi'(x) dx = - \int_{\mathbb{R}} \text{sgn}(x)\varphi(x) dx,$$

so the weak derivative of f is $\text{sgn}(x)$, which belongs to $L^\infty(\mathbb{R}^n) \subset L^1_{\text{loc}}(\mathbb{R})$.

Remark 5.1.7. If $f \in C^1(\mathbb{R}^n)$, then its weak derivative coincides with the classical one. Thus the weak derivative extends the classical notion without contradiction.

In this text we are only going to consider Sobolev spaces in the entire space $\mathbb{R}^n$. One can also define Sobolev spaces on bounded domains $\Omega \subset \mathbb{R}^n$, but this requires additional work.

5.2 Integer order Sobolev spaces

Definition 5.2.1 (Integer-order Sobolev spaces). Let $k \in \mathbb{N}$ and $1 \leq p \leq \infty$. Then the classical Sobolev space

$$W^{k,p}(\mathbb{R}^n) := \{u \in L^p \mid D^\beta u \in L^p \ \forall |\beta| \leq k\}$$

is the space of L^p -functions, whose weak derivatives also belong to L^p . The space is normed, with the norm given by

$$\|f\|_{W^{k,p}} := \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p}^p \right)^{\frac{1}{p}}.$$

If $p = \infty$ we define

$$\|f\|_{W^{k,\infty}} := \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^\infty}.$$

5.3 Bessel potential spaces

Definition 5.3.1 (Bessel potential space). For $s \in \mathbb{R}$ and $1 \leq p \leq \infty$, the *Bessel potential space* is defined as

$$H^{s,p}(\mathbb{R}^n) := \left\{ u \in \mathcal{S}'(\mathbb{R}^n) \mid \mathcal{F}^{-1}((1 + |\xi|^2)^{s/2} \hat{u}(\xi)) \in L^p(\mathbb{R}^n) \right\}.$$

Thus, $H^{s,p}(\mathbb{R}^n)$ consists of all tempered distributions f such that

$$\mathcal{F}^{-1}((1 + |\xi|^2)^{s/2} \hat{f}(\xi)) \in L^p(\mathbb{R}^n),$$

with norm

$$\|f\|_{H^{s,p}} := \left\| \mathcal{F}^{-1}((1 + |\xi|^2)^{s/2} \hat{f}(\xi)) \right\|_{L^p}.$$

Remark 5.3.2. We will often write the *Bessel potential*

$$J^s f := \mathcal{F}^{-1}((1 + |\xi|^2)^{s/2} \hat{f}(\xi))$$

as the Fourier multiplier operator

$$(1 - \Delta)^{\frac{s}{2}} f := \langle D \rangle^s f := J^s f,$$

where $\langle \xi \rangle := (1 + |\xi|^2)^{\frac{1}{2}}$ is called the Japanese bracket, and we recall $D^\alpha = (i\partial)^\alpha$. The $H^{s,p}$ -norm can then be written as

$$\|u\|_{H^{s,p}} = \|(1 - \Delta)^{s/2} u\|_{L^p} = \|\langle D \rangle^s u\|_{L^p}.$$

All of the above notations are common in the literature; $(1 - \Delta)^{s/2}$ is a common *pseudodifferential operator*, the operator J^s appears often in theory of PDEs and harmonic analysis, and $\langle D \rangle^s$ in microlocal analysis and semiclassical PDEs.

Beware that some authors define the Bessel potential with different sign-conventions. Always check what notation the author is using.

The intuition is simply that the Bessel potential space consists of those functions, whose Fourier transform happens to satisfy the above property in the L^p space. It's not a mystical thing; in fact

$$W^{k,p} = H^{k,p}, \quad \text{if } k \geq 0 \text{ is an integer.}$$

We will prove this fact in Theorem 5.3.8. It is for this reason that we define the *fractional Sobolev spaces* by

$$W^{s,p} := H^{s,p}, \quad \text{when } s \in \mathbb{R}_+ \setminus \mathbb{N}.$$

Remark 5.3.3. There are many possible definitions of Sobolev spaces. For example, the fractional Sobolev spaces $\widetilde{W}_p^s$ could be defined using the Gagliardo seminorm (such spaces are also called Gagliardo spaces or Slobodeckij spaces). This space is *not* the same space as $H^{s,p}$. In fact, it can be shown that $H^{s,p} \subset \widetilde{W}^{s,p}$ when $p \geq 2$ and $\widetilde{W}^{s,p} \subset H^{s,p}$, when $p \leq 2$ (for proof, see [6, Chapter V, Theorem 5]). When reading mathematical texts involving Sobolev spaces, take care to check which definition the author is using.

Already in Theorem 2.1.3 we have seen that, in the context of Fourier transforms, differentiation in the spatial domain equals multiplication by the frequency in the Fourier domain. Similarly, translations become multiplication by the exponential. This property is useful and appears over and over again, which leads us to

Definition 5.3.4 (Fourier multiplier operators). Let $m : \mathbb{R}^n \rightarrow \mathbb{C}$ be a measurable function. The *Fourier multiplier operator* T_m is defined by the equation

$$\widehat{T_m f} = m(\xi) \widehat{f}(\xi).$$

In general, the mapping properties of Fourier multiplier operators are difficult to obtain.

Theorem 5.3.5. *If $m \in L^\infty(\mathbb{R}^n)$, then the corresponding Fourier multiplier operator $T_m : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$.*

Proof. Plancherel's theorem. □

If m is smooth enough with a decay fast enough, we have the useful

Theorem 5.3.6 (Mikhlin multiplier theorem). *Let $1 < q < \infty$ and let $m \in C^N(\mathbb{R}^n \setminus \{0\})$,*

$$|\partial^\beta m(\xi)| \leq C_\beta (1 + |\xi|^2)^{-\frac{|\beta|}{2}}, \quad \forall |\beta| \leq \left\lfloor \frac{n}{2} \right\rfloor + 1, \quad \xi \neq 0.$$

Then the Fourier multiplier operator

$$T_m f := \mathcal{F}^{-1}(m(\xi)\hat{f}(\xi))$$

extends to a bounded linear operator on $L^q(\mathbb{R}^n)$ and

$$\|T_m f\|_{L^q} \leq C_{n,q,m} \|f\|_{L^q}.$$

Proof. A proof can be found in [1, Theorem 6.2.7]. □

A multiplier m satisfying the assumptions of Theorem 5.3.6 is called a *Mikhlin multiplier* or the *symbol* of T_m .

Remark 5.3.7. The following proof is adapted from Mikko Salo's lectures on Function Spaces, 2008.

Theorem 5.3.8. *Let $1 < p < \infty$ and $k \in \mathbb{Z}_+$. Then $W^{k,p} = H^{k,p}$ with equivalent norms.*

Proof. 1° $H^{k,p} \subset W^{k,p}$. Let $u \in H^{k,p}(\mathbb{R}^n)$ and let $|\alpha| \leq k$. Our task is to show that $D^\alpha u \in L^p(\mathbb{R}^n)$ for all $|\alpha| \leq k$. We can write

$$D^\alpha u = m(D)(\langle D \rangle^k u), \quad m(\xi) := \frac{\xi^\alpha}{\langle \xi \rangle^k}.$$

Now m is a Mikhlin multiplier and $m(D) : L^p(\mathbb{R}^n) \rightarrow L^p(\mathbb{R}^n)$. Since $\langle D \rangle^k u \in L^p(\mathbb{R}^n)$ we have $D^\alpha u = m(D)(\langle D \rangle^k u) \in L^p(\mathbb{R}^n)$ for any $|\alpha| \leq k$, and Mikhlin implies

$$\|u\|_{W^{k,p}} \leq C \|u\|_{H^{k,p}}.$$

2° $W^{k,p} \subset H^{k,p}$. This is the more challenging direction. Let $u \in W^{k,p}(\mathbb{R}^n)$. We need to show that $\langle D \rangle^k u \in L^p(\mathbb{R}^n)$ given that $D^\alpha u \in L^p(\mathbb{R}^n)$ for all $|\alpha| \leq k$.

Let now $\varphi \in C^\infty(\mathbb{R})$ be such that $0 \leq \varphi \leq 1$, $\varphi(s) = 0$ if $|s| \leq 1$ and $\varphi(s) = 1$ if $|s| \geq 2$. (Note that φ is defined on the real axis, and is not compactly supported.) We next do nothing, and write

$$\langle D \rangle^k u = \mathcal{F}^{-1}((1 + |\xi|^2)^{\frac{k}{2}} \hat{u}) = \mathcal{F}^{-1} \left(m_0(\xi) \left(1 + \sum_{j=1}^n \varphi(\xi_j) |\xi_j|^k \right) \hat{u} \right)$$

where

$$m_0(\xi) = \frac{(1 + |\xi|^2)^{\frac{k}{2}}}{1 + \sum_{j=1}^n \varphi(\xi_j) |\xi_j|^k}.$$

Now, the goal is to write down derivatives of u . By linearity of the Fourier transform

$$\langle D \rangle^k u = \mathcal{F}^{-1}(m_1(\xi)\widehat{u}) + \sum_{j=1}^n m_0(D)m_j(D_j)|\xi_j|^k D_j^k u,$$

where the multipliers

$$m_j(\xi_j) = \varphi(\xi_j)|\xi_j|^k \xi_j^{-k}$$

(recalling that $\varphi(\xi_j) \equiv 0$ in a neighbourhood of $\xi_j = 0$, so we are not dividing by zero).

Now, it is not difficult to verify that the multipliers $m_0, m_j, j = 1, \dots, n$ are Mihlin multipliers in $L^p(\mathbb{R}^n)$ and $L^p(\mathbb{R})$, respectively. Indeed, m_0 is smooth near $\xi = 0$, because $\varphi = 0$ near 0. Moreover,

$$(1 + |\xi|^2)^{\frac{k}{2}} \leq (1 + n \max_j |\xi_j|^2)^{\frac{k}{2}} \leq C(1 + \max_j |\xi_j|^k) \leq C(1 + \sum_{j=1}^n |\xi_j|^k),$$

proving that m_0 is bounded. For the derivatives we directly have that

$$|D^\beta m_0(\xi)| \leq \frac{C}{(1 + |\xi|^2)^{-\frac{|\beta|}{2}}},$$

showing that m_0 is Mihlin multiplier. This shows that

$$\|\langle D \rangle^k u\|_{L^p} \leq C \left(\|u\|_{L^p} + \sum_{j=1}^n \|m_j(D_j)(D_j^k u)\|_{L^p} \right)$$

Finally, the functions m_j satisfy Mihlin conditions directly (they are obviously bounded and $\partial_j m_j(\xi_j) = \varphi'(\xi_j)\xi_j/|\xi_j| = 0$, when $|\xi_j| \geq 2$). Hence

$$\begin{aligned} \|m_j(D_j)(D_j^k u)\|_{L^p(\mathbb{R}^n)} &= \|m_j(D_j)(D_j^k u)\|_{L^p(\mathbb{R})} \|m_j(D_j)(D_j^k u)\|_{L^p(\mathbb{R}^{n-1})} \\ &\leq C \|D_j^k u\|_{L^p(\mathbb{R})} \|D_j^k u\|_{L^p(\mathbb{R}^{n-1})} = \|D_j^k u\|_{L^p(\mathbb{R}^n)} < \infty. \end{aligned}$$

This holds for all $j = 1, \dots, n$, implying $\langle D \rangle^k u \in L^p(\mathbb{R}^n)$. Summing over $j = 1, \dots, n$ yields

$$\|u\|_{H^{k,p}} \leq C \|u\|_{W^{k,p}}.$$

□

Exercise 5.3.9. Assume that $k \geq 0$ is an **even** integer and $1 < p < \infty$. Show by a direct calculation that there is a constant $C_1 > 0$, such that $\|f\|_{H^{k,p}} \leq C_1 \|f\|_{W^{k,p}}$, without resorting to Mihlin's theorem.

Exercise 5.3.10. Let $1 < p < \infty$ and $t \geq s$. Prove that

$$H^{t,p}(\mathbb{R}^n) \subset H^{s,p}(\mathbb{R}^n).$$

Hint: Write

$$(1 + |\xi|^2)^{s/2} = (1 + |\xi|^2)^{(s-t)/2} (1 + |\xi|^2)^{t/2}$$

and use Mihlin's multiplier theorem.

The monotonicity of Bessel potential spaces

$$H^{t,p} \subset H^{s,p}, \quad t \geq s,$$

is immediate when $p = 2$ by Plancherel's theorem. For $p \neq 2$, however, this is a nontrivial statement about L^p -boundedness of Fourier multipliers. The boundedness of Bessel potentials, Riesz transforms, and related operators forms a cornerstone of Calderón–Zygmund theory.

5.4 The Sobolev space $H^s(\mathbb{R}^n)$

Lemma 5.4.1. *Let $s \in \mathbb{R}$. Then the space $H^{s,2}(\mathbb{R}^n)$ is defined by the norm*

$$\|f\|_{H^{s,2}} := \|(1 + |\xi|^2)^{\frac{s}{2}} \widehat{f}\|_{L^2}.$$

Proof. This is just Plancherel's theorem

$$\|f\|_{H^{s,2}} = \|\mathcal{F}^{-1}((1 + |\xi|^2)^{\frac{s}{2}} \widehat{f})\|_{L^2} = \|(1 + |\xi|^2)^{\frac{s}{2}} \widehat{f}\|_{L^2}.$$

□

In the case $p = 2$ we simply write

$$H^s := H^{s,2}.$$

This is perhaps the most important Sobolev space, in the sense that it is a Hilbert space. Fourier analysis is rather simple in the L^2 -based case due to Plancherel's theorem and many orthogonality arguments.

Theorem 5.4.2. *If $s > k + n/2$, then*

$$H^s(\mathbb{R}^n) \subset C_0^k(\mathbb{R}^n).$$

Proof. Hint: Verify that $D^\alpha f$ is the Fourier transform of some function g belonging to L^1 , when $s > |\alpha| + n/2$. Then, use the Riemann–Lebesgue lemma (Lemma 3.0.1) to verify that derivatives of f are (uniformly) continuous and vanish at infinity. □

Exercise 5.4.3. Show that the characteristic function $\chi_{[0,1]} \in H^s(\mathbb{R})$ if and only if $s < \frac{1}{2}$.

Exercise 5.4.4. Show that the delta-distribution $\delta \in H^s(\mathbb{R}^n)$ if and only if $s < -\frac{n}{2}$.

Theorem 5.4.5 (Uncertainty principle). *Let $f \in L^2(\mathbb{R})$ be real-valued with $xf, f' \in L^2(\mathbb{R})$ and $\|f\|_{L^2} = 1$. Define the variances*

$$\sigma_x^2 = \int_{\mathbb{R}} x^2 f(x)^2 dx, \quad \sigma_\xi^2 = \int_{\mathbb{R}} \xi^2 |\widehat{f}(\xi)|^2 d\xi.$$

Then

$$\sigma_x \sigma_\xi \geq \frac{1}{2},$$

and equality holds precisely for Gaussians.

Proof. By Plancherel and $\widehat{f'}(\xi) = i\xi\widehat{f}(\xi)$ we have

$$\|f'\|_{L^2}^2 = \int_{\mathbb{R}} |f'(x)|^2 dx = \int_{\mathbb{R}} |\widehat{f'}(\xi)|^2 d\xi = \int_{\mathbb{R}} \xi^2 |\widehat{f}(\xi)|^2 d\xi = \sigma_{\xi}^2.$$

By Cauchy–Schwarz,

$$|\langle f', xf \rangle|^2 \leq \|f'\|_{L^2}^2 \|xf\|_{L^2}^2 = \sigma_{\xi}^2 \sigma_x^2.$$

Since $f, f' \in L^2(\mathbb{R})$, we have $f \in H^1(\mathbb{R})$; by the one-dimensional Sobolev embedding an H^1 -representative is continuous and tends to 0 at $\pm\infty$. Hence we may integrate by parts

$$\langle f', xf \rangle = \int_{\mathbb{R}} f'(x) xf(x) dx = -\frac{1}{2} \int_{\mathbb{R}} f(x)^2 dx = -\frac{1}{2},$$

using the fact that the boundary terms vanish. Therefore

$$\frac{1}{4} = \left(\frac{1}{2}\right)^2 = |\langle f', xf \rangle|^2 \leq \sigma_{\xi}^2 \sigma_x^2,$$

and taking square roots yields

$$\sigma_x \sigma_{\xi} \geq \frac{1}{2}.$$

Equality in Cauchy–Schwarz occurs iff f' and xf are linearly dependent in L^2 , i.e. $f' = \lambda xf$ a.e. for some (real) constant λ . Solving the ODE gives

$$f(x) = A \exp\left(\frac{\lambda}{2}x^2\right).$$

Integrability forces $\lambda < 0$, so f is (up to scaling) a Gaussian. Conversely, such Gaussians attain equality. $\square$

Remark 5.4.6 (Hardy’s uncertainty principle). If f and $\widehat{f}$ both decay faster than any Gaussian, then $f = 0$. This shows that the Gaussian is the most localized function in the joint time-frequency sense.

Remark 5.4.7. Recall that $\widehat{\delta} \sim 1$. This can be interpreted as stating, that maximal localization (unit mass at a single point) contains equal amount of all possible frequencies.

5.5 Sobolev embedding theorem

Before going to the main result about Sobolev spaces, we require certain mapping properties of an integral operator.

Definition 5.5.1 (Riesz potential). Let $0 < \alpha < n$ and $f \in L^p(\mathbb{R}^n)$ for some $1 \leq p < \infty$. The *Riesz potential* of order α is defined by

$$I_{\alpha}f(x) := 2^{-s}\pi^{-\frac{n}{s}} \frac{\Gamma\left(\frac{n-\alpha}{2}\right)}{\Gamma\left(\frac{\alpha}{2}\right)} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-\alpha}} dy,$$

where Γ is the gamma-function.

It can be shown that the Riesz potential is, in fact, a Fourier multiplier operator

$$I_\alpha f = C_{n,\alpha} \mathcal{F}^{-1}(|\xi|^{-\alpha} \widehat{f})$$

with the Fourier symbol $|\xi|^{-\alpha}$. Its mapping properties are quite challenging to prove, and we refer to Grafakos [2, Theorem 6.1.3] and Stein [6, Chapter V.1.2, Theorem 1]. The main result for the Riesz potential is the following

Theorem 5.5.2 (Hardy–Littlewood–Sobolev inequality). *Let $0 < \alpha < n$ and $1 < p < q < \infty$ satisfy*

$$\frac{1}{q} = \frac{1}{p} - \frac{\alpha}{n}. \quad (5.1)$$

Then the Riesz potential I_α extends to a bounded linear operator

$$I_\alpha : L^p(\mathbb{R}^n) \longrightarrow L^q(\mathbb{R}^n),$$

i.e. there exists a constant $C = C(n, p, \alpha)$ such that

$$\|I_\alpha f\|_{L^q} \leq C \|f\|_{L^p} \quad \text{for all } f \in L^p(\mathbb{R}^n).$$

Proof. The proof uses the mapping properties of the Hardy–Littlewood maximal function and Vitali’s covering lemma, boiling down to the Marcinkiewicz interpolation theorem. We omit the proof. $\square$

Exercise 5.5.3. Verify, using definition of the Riesz potential I_α , that if $\|I_\alpha f\|_{L^q} \leq C \|f\|_{L^p}$, for all $f \in L^p(\mathbb{R}^n)$, then p and q must necessarily be related through (5.1). Hint: apply the dilation σ_h to f and do changes of variables.

Definition 5.5.4. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed vector spaces. We say that X is *continuously embedded in Y* and denote $X \hookrightarrow Y$, if $X \subset Y$ and the embedding operator $i : X \rightarrow Y$ mapping $i(x) = x$ for all $x \in X$ is continuous. Linear operators are continuous if and only if they are bounded, so the above is equivalent with the fact that there exists a constant $C > 0$ such that $\|x\|_Y \leq C \|x\|_X$ for all $x \in X$.

Theorem 5.5.5 (Sobolev embedding theorem). *Let $1 < p < q < \infty$, and $s, r \in \mathbb{R}$ satisfy*

$$\frac{1}{p} - \frac{s}{n} = \frac{1}{q} - \frac{r}{n}. \quad (5.2)$$

Set $\alpha := s - r$. Assume that $0 < \alpha < n$ and $1 < p < \frac{n}{\alpha}$. Then there is a continuous embedding

$$H^{s,p}(\mathbb{R}^n) \hookrightarrow H^{r,q}(\mathbb{R}^n).$$

Remark 5.5.6. The equality (5.2) can be replaced by the inequality

$$\frac{1}{p} - \frac{s}{n} \leq \frac{1}{q} - \frac{r}{n},$$

which is often easier to verify.

Remark 5.5.7. A common use of the Sobolev embedding theorem is as follows: Given p, q, r as in the theorem, determine the minimal $s = s^*$ such that the embedding $H^{s,p} \hookrightarrow H^{r,q}$ holds. Then, by the monotonicity of Sobolev spaces $H^{s',p} \subset H^{s,p}$ for $s' \geq s$, it follows immediately that $H^{s',p} \hookrightarrow H^{r,q}$ for all $s' \geq s$.

Exercise 5.5.8. Show that $H^1(\mathbb{R}^2) \hookrightarrow L^q(\mathbb{R}^2)$ for all $2 \leq q < \infty$, but $H^1(\mathbb{R}^2) \not\hookrightarrow L^\infty(\mathbb{R}^2)$. This shows that $q = \infty$ is not allowed in Theorem 5.5.5.

Remark 5.5.9. In 1D we can prove more: We have the continuous embedding

$$W^{k+1,p}(\mathbb{R}) \hookrightarrow C^{k,1/q}(\mathbb{R})$$

for integers $k \geq 0$ and all $1 < p \leq \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$. The proof is not so difficult and we will cover it in Exercise 6. Here the space $C^{k,1/2}(\mathbb{R})$ consists of C^k -functions whose k th derivative is $1/q$ -Hölder continuous. This does not, in general, hold for $n \geq 2$.

Proof of Theorem 5.5.5. Let $f \in H^{s,p}(\mathbb{R}^n)$ and define

$$g := \mathcal{F}^{-1}((1 + |\xi|^2)^{s/2} \widehat{f}) = (I - \Delta)^{s/2} f.$$

Then, by definition,

$$g \in L^p(\mathbb{R}^n), \quad \text{and} \quad \|g\|_{L^p} = \|f\|_{H^{s,p}}.$$

Now, applying the Bessel potential J^r to f , we see that

$$\mathcal{F}(J^r f) = (1 + |\xi|^2)^{\frac{r}{2}} \widehat{f}(\xi).$$

Thus,

$$(1 - \Delta)^{r/2} f(\xi) = (1 + |\xi|^2)^{r/2} \widehat{f}(\xi) = (1 + |\xi|^2)^{(r-s)/2} \widehat{g}(\xi).$$

Therefore $J^r f$ is given by $T_m g$ for some $g \in L^p(\mathbb{R}^n)$ as

$$(1 - \Delta)^{r/2} f = T_m g, \quad m(\xi) := (1 + |\xi|^2)^{-\alpha/2}, \quad \alpha := s - r > 0.$$

We next factorize the multiplier m into a Mihlin multiplier and a Riesz potential. Precisely, write

$$m(\xi) = m_1(\xi) |\xi|^{-\alpha}, \quad m_1(\xi) := (1 + |\xi|^2)^{-\alpha/2} |\xi|^\alpha.$$

Hence

$$(1 - \Delta)^{r/2} f = T_{m_1} \circ I_\alpha g = T_{m_1}(I_\alpha g),$$

where I_α is the Riesz potential of order α , with Fourier multiplier $|\xi|^{-\alpha}$ and T_{m_1} the Mihlin multiplier operator.

We verify that the Hardy-Littlewood-Sobolev inequality applies to I_α . Since $\alpha = s - r \in (0, n)$ and

$$\frac{1}{q} = \frac{1}{p} - \frac{\alpha}{n},$$

the assumptions of the Hardy–Littlewood–Sobolev inequality are satisfied, and therefore

$$\|I_\alpha g\|_{L^q} \leq C_{n,q,p} \|g\|_{L^p}. \quad (5.3)$$

Next, we claim $m_1(\xi)$ is a Mikhlin multiplier. Indeed, m_1 is radial:

$$m_1(\xi) = \psi(|\xi|), \quad \psi(t) = t^\alpha (1 + t^2)^{-\alpha/2}.$$

For every integer $k \geq 0$, $\psi^{(k)}$ is a finite sum of terms of the form

$$t^{\alpha-\ell+r} (1 + t^2)^{-\alpha/2-m}.$$

From this one checks that

$$\sup_{t>0} |t^k \psi^{(k)}(t)| < \infty.$$

Hence for each multi-index β with $|\beta| \leq N$ (where $N > \frac{n}{2}$),

$$|\partial^\beta m_1(\xi)| \leq C_\beta |\xi|^{-|\beta|}, \quad \xi \neq 0.$$

Therefore T_{m_1} is bounded on L^q , $1 < q < \infty$, by the Mikhlin multiplier theorem:

$$\|T_{m_1} h\|_{L^q} \leq C_{m_1,q} \|h\|_{L^q}. \quad (5.4)$$

Combining the estimates (5.3) and (5.4), we have

$$\|(1 - \Delta)^{r/2} f\|_{L^q} = \|T_{m_1} \circ I_\alpha g\|_{L^q} \leq C_{m_1,q} \|I_\alpha g\|_{L^q} \leq C_{m_1,q} C_{n,q,p} \|g\|_{L^p}.$$

Recalling $\|g\|_{L^p} = \|f\|_{H^{s,p}}$, we conclude

$$\|f\|_{H^{r,q}} \leq C \|f\|_{H^{s,p}}.$$

Thus $H^{s,p} \hookrightarrow H^{r,q}$, as required. $\square$

Remark 5.5.10. A clean proof can also be found, for example, in the book of Grafakos [2, Chapter 6.2], where the continuous embedding

$$H^{s,p} \hookrightarrow L^q$$

for $1 < p < \infty$, $0 < s < n/p$, $1/p - 1/q = s/n$ is given. This result directly extends to the statement of Theorem 5.5.5 by using the fact that $H^{s,p} \hookrightarrow H^{r,p}$, whenever $s > r$. (Let $J^s u = \langle D \rangle^s u$ be the Bessel potential of degree s , and use Grafakos' result for the space $H^{s-r,p}$ to put $J^r u$ into L^q for $u \in H^{s,p}$, then observe that $J^r u \in L^q$ means $u \in H^{r,q}$.)

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